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Build Direction Effects on Additively Manufactured Channels

With the advance of direct metal laser sintering (DMLS), also generically referred to as additive manufacturing (AM), novel geometric features of internal channels for gas turbine cooling can be achieved beyond those features using traditional manufacturing techniques. There are many variables, however, in the DMLS process that affect the final quality of the part. Of most interest to gas turbine heat transfer designers are the roughness levels and tolerance levels that can be held for the internal channels. This study investigates the effect of DMLS build direction and channel shape on the pressure loss and heat transfer measurements of small-scale channels. Results indicate that differences in pressure loss occur between the test cases with differing channel shapes and build directions, while little change is measured in heat transfer performance.

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Introduction

Recent advances in AM, also known as 3D printing, have allowed complex geometric designs to be created with many metal alloys. Selective laser melting (SLM), electron beam melting, and selective laser sintering (SLS) are all technologies capable of manufacturing metal parts. Specifically, DMLS, a type of SLS, has been the frontrunner in creating production grade parts. Sintering methods are of interest to the aerospace community because sintering allows parts to be manufactured from high-temperature alloys, such as Inconel; however, because the metal powder is sintered and not melted, porosity and microstructure can differ from a cast part of the same material. This uncertainty in material properties makes DMLS unfavorable for use in high stress turbine parts, such as blades, but better suited for high-temperature parts in nonload-bearing applications.

With these limitations in mind, one possible application of metal AM is static hot section parts with complex cooling technologies. To combat increasing gas turbine temperatures, high efficiency cooling designs are required in numerous locations. Currently, the complexity of these cooling designs is limited by traditional manufacturing methods. Using DMLS, designers now have the opportunity to create cooling geometries that would be impossible to achieve with traditional methods.

The DMLS process introduces a number of variables that affect the final quality of the part. Laser scan speed, layer thickness, support structures, and part orientation are among many factors that contribute to the final geometric tolerances and surface roughness.

Of particular interest to a gas turbine heat transfer designer are the tolerance and surface roughness of internal channels. These two channel properties are highly affected by build direction. Additionally, current commercial surface treatment processes are not capable of smoothing internal channels at this small scale. Therefore, the goal of this study is to evaluate the effect of DMLS build direction on the pressure loss and heat transfer performance of small (mini) channels. The intention of this study was not to evaluate the effect of various DMLS process parameters on the overall part quality. Instead, parameters recommended by the manufacturer were used for the test coupons, which were constructed in-house with state-of-the-art AM equipment.

Review of Literature

Since DMLS is a relatively new technology, most open literature in this area focuses on the fundamentals of the laser sintering process and evaluating the materials produced by this method by looking at microstructure and mechanical properties. While this fundamental research has been essential for developing the process, few studies have investigated the application of DMLS to actual component quality.

Some of these fundamental studies on DMLS parts have quantified the surface roughness. Simchi et al. [1] studied a simple iron-based DMLS part and reported surface roughness of $10 < R_a < 12 \mu\text{m}$. Khaing et al. [2] reported roughness between $12 < R_a < 16 \mu\text{m}$ for a nickel-bronze-copper DMLS part. In addition to surface roughness, Khaing et al. evaluated the geometric tolerance of the test specimen. This test specimen featured a cylinder, a hole, cones, rectangular ribs, and small features. The dimensional errors in the part resulting from the DMLS process varied from 0.003 mm to 0.082 mm. Specifically, significant inaccuracy

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of cylindrical features was found; deviations from the design diameter were 0.025–0.34 mm.

Expanding on these preliminary findings, a number of studies have been published investigating the effect of process variables on the final part quality. Song and Koenig [3], Calignano et al. [4], and Senthilkumaran et al. [5] investigated the effect of various DMLS process parameters (laser scan speed, laser power, and hatching distance) on the surface roughness. In addition to investigating these parameters, Delgado et al. [6] studied the effect of build direction on surface roughness and dimensional tolerance for stainless steel DMLS parts. While this study only investigated two build directions, the authors found that the build direction had a much larger effect on surface roughness and dimensional tolerance than the aforementioned process parameters. Other studies [7,8] have investigated the effect of build direction on mechanical properties of SLM and DMLS parts. However, no build direction studies have examined the effects on internal channel geometry.

Ventola et al. [9] investigated the effect of build direction on surface roughness, with similar findings to Delgado et al. [6]. In addition to roughness measurements, the authors also performed external flow and heat transfer measurements. The heat transfer data agreed with external flow correlations for the measured surface roughness levels. While this was a thorough study, internal flow of DMLS parts is also of interest to designers.

Wong et al. [10] was one of the few studies to examine heat transfer and pressure loss through additively manufactured heat exchangers. Three different pin fin geometries and a lattice design were fabricated with SLM, and then tested for heat transfer and pressure loss. Friction factor for a rectangular fin design agreed with the correlation for rectangular offset heat exchangers proposed by Manglik and Bergles [11]. However, this study did not construct the entire heat exchanger with SLM. Large-scale turbulators were built separately and then installed in an existing channel for testing. Therefore, the effects of using metal AM for embedded cooling channels are still unexplored.

As Bunker [12] alluded, AM will play a role in moving from macro- to microcooling. The open design space that AM provides is extremely desirable. In fact, some recent studies by Helmer [13] and Kinel et al. [14] have utilized AM, DMLS, and SLM, respectively, to create test specimens for gas turbine cooling research. The focus of these studies, however, was on measurement techniques. The effect of using AM to create the test articles was not explored.

At the small-scale associated with gas turbine cooling, the surface roughness and dimensional tolerance of internal channels are critical. To the best of our knowledge, there have been no studies published that investigate internal feature quality of DMLS parts, especially for minichannels. Therefore, the goal of this study is to investigate the effect of build direction on the quality of internal minichannels. Specifically, metrics useful to a gas turbine designer, such as friction factor and heat transfer, will be used to evaluate build direction effects.

DMLS Coupon Description

To examine the build direction effects on surface roughness, geometric tolerance, and ultimately the pressure loss and heat transfer, different test coupons were designed and manufactured in-house. As shown in Fig. 1, the overall size of the coupons was 25.4 mm long $\times$ 25.4 mm wide $\times$ 3.048 mm high. Each coupon had 15 channels spaced 2.5 diameters apart, with a designed hydraulic diameter of $D_h = 508 \mu\text{m}$. The material used for manufacturing the coupons was Inconel 718.

The three different build directions studied are shown in Figs. 2(a)–2(c) including horizontal, diagonal, and vertical. Each build orientation was defined by the position of the channel axes relative to the build plate. The channel axes were parallel to the build plate in the horizontal build direction, 45 deg to the build plate in the diagonal build direction, and perpendicular to the build plate in the vertical build direction.

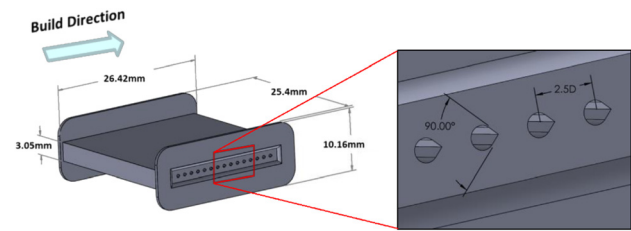


Fig. 1 CAD model of coupon showing overall dimensions and channel shape detail for the teardrop design

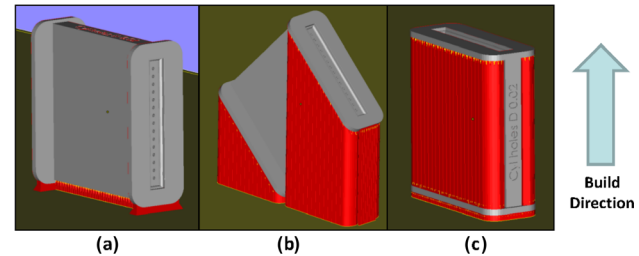


Fig. 2 Support structures used for the (a) horizontal, (b) 45 deg, and (c) vertical build directions

Cylindrical-shaped channels were chosen as the baseline to investigate build direction effects since it is a common design. In addition to the cylindrical channels, channels with diamond and teardrop shapes were also investigated. The design intent for the diamond and teardrop shapes was to create a channel shape that would not collapse when built in the horizontal build direction. These alternate channel shapes were designed to match the cylindrical channel $D_h = 508 \mu\text{m}$. A computer-aided design (CAD) model of the teardrop coupon can be seen in Fig. 1, while all of the designed channel shapes are shown in Fig. 3.

Building these coupons required support structures to be generated. Most AM methods require structures to support overhanging features. Specific to the DMLS process with Inconel 718, the minimum recommended unsupported angle is 40 deg. Therefore, any feature of the test coupons that was less than 40 deg to horizontal required supports. Using this criteria, the upper surface of the horizontal cylindrical channels required supports; however, it would be difficult to remove these supports after the build was completed. As such, the horizontal surfaces of the channels were intentionally left unsupported to replicate situations where internal channels require supports, but cannot be accessed for support removal.

Resulting support structures are shown in Fig. 2. In addition to surfaces below 40 deg, supports were placed on other features, such as the coupon flanges required for the testing facility, to anchor them to the build plate to prevent distortion. For the study presented in this paper, the support structures were generated using a commercial stereolithography file editing and build preparation software.

A state-of-the-art DMLS machine was used to manufacture the coupons. Three duplicates of each test case were manufactured from the Inconel 718 powder for a total of 15 coupons. These coupons were arranged in rows on the build plate. To complete the build, the machine parameters in Table 1 were used. A previously built qualification block was used to calculate the material scalings and beam offset. These calculations were carried out per the manufacturer's specifications [15]; no advanced process parameter development was used for this study. Additionally, since the goal of this study was to investigate build direction, the process parameters were kept constant across all coupons. The total time for the build was 53 hrs, which included a total of 76 coupons; however, not all coupons built are presented in this study.

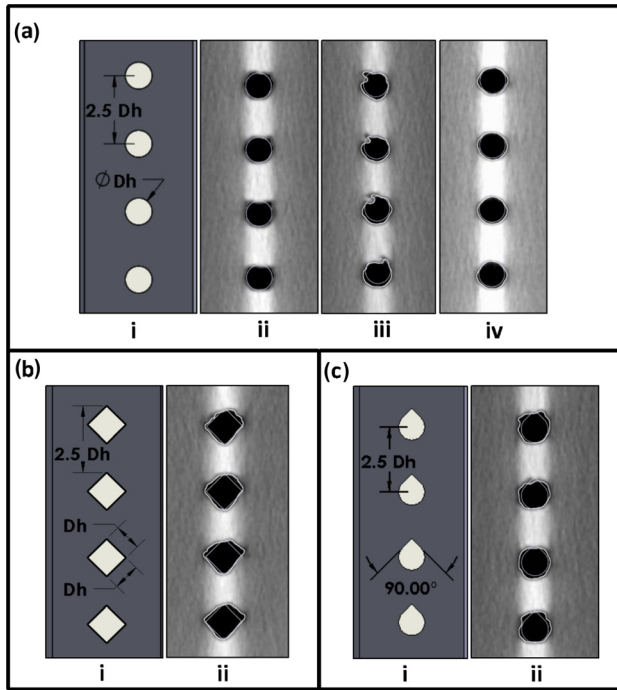


Fig. 3 Cross sections showing design compared to CT scan of actual parts built: (a) cylindrical—(i) design intent, (ii) horizontal build, (iii) diagonal build, and (iv) vertical build, (b) diamond—(i) design intent and (ii) horizontal build, and (c) teardrop—(i) design intent and (ii) horizontal build. Build directions are defined in Fig. 2.

Table 1 DMLS machine settings [15]

Parameter	Value
Material setting	IN718 040 performance
Material scaling X	0.045%
Material scaling Y	0.16%
Beam offset	0.11 mm
Layer thickness	0.04 mm

Heat treatment of the coupons was needed to remove residual stresses from the manufacturing process. After the heat treatment, a wire electrical discharge machining (EDM) was used to cut the coupons from the build plate. Once removed from the plate, remaining support structures were removed using manual tools.

To inspect the internal features of the coupons, computed X-ray tomography (CT scan) was used as a nondestructive evaluation tool. As was shown in Fig. 3, the CT scan images showed significant differences between the intended channel shape and the resulting channel shapes from the DMLS. These differences in channel shapes will be characterized later in the paper. Qualitatively, however, the vertical build direction shows the least degradation in the channel shape, which was expected since all channel surfaces were supported. The channels built horizontally, however, have a collapsed upper surface. This collapsed upper surface was mitigated with the teardrop- and diamond-shaped channels. In both cases, the two upper sides forming the 90 deg angle fused together. The resulting cross section of the teardrop-shaped channel was found to best replicate the circular channel.

Geometric Characterization

Determining the actual channel dimensions and roughness was essential for understanding the differences in flow behavior between build orientations and coupons. To obtain these

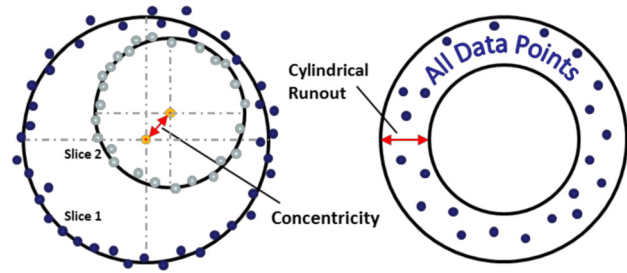


Fig. 4 Diagram defining concentricity and cylindrical runout

measurements, the CT scanner was used to generate a 3D reconstruction of each coupon, with a resolution (i.e., voxel size) of 35 μm . The surface location for each coupon was extracted by comparing grayscale values using commercial imaging software. Using a local comparison of these gray values, accuracies in surface determination up to 1/10th of the voxel size can be achieved [16]. Thus, the surface determined with the CT scan is accurate to within 3.5 μm .

To perform further analysis on the CT scan data, the determined surface was exported as a three-dimensional point cloud. The number of points defining each channel was monitored to ensure that the spatial resolution of the point cloud matched the expected spatial resolution of the surface determination. In most cases, the point cloud was nominally 500 thousand points per channel. An in-house code was used to analyze the point clouds defining each channel separately.

To quantify the hydraulic diameter from the CT scan data, two different methods were used. The first method involved finding the channel axis and then averaging the distance of all points from this axis. The resulting value was an average radius, which was then doubled to obtain an average hydraulic diameter. This method was limited to the vertically built channels which were closest to a perfect cylinder as shown in Fig. 3(a-iv). For the other channels which deviated from the cylindrical shape, a different analysis was used to better capture the hydraulic diameter. The point cloud was first cut into 500 equal slices in the axial direction. Points located within each slice were all considered to lie on the same plane. Each plane was then split and fourth-order polynomials were fit to the data on each half. Computing the length and integral of these polynomials gave the perimeter and area, respectively. A total perimeter and cross-sectional area for a slice were then determined by summing each value for the two slice halves. The mean was then taken over all slices for a given channel. To find the total hydraulic diameter of the coupon, the mean areas and perimeters for each channel were summed as shown in the following equation:

$$D_h = 4 \frac{\sum_{m=1}^{15} A_c}{\sum_{m=1}^{15} P} \quad (1)$$

The 3D point cloud was also used to calculate cylindrical runout and concentricity tolerances. Cylindrical runout was calculated by determining the distance between the largest circle that could inscribe the data and the smallest circle that could circumscribe the data, as illustrated in Fig. 4. A perfect cylindrical shape has zero cylindrical runout. Concentricity was calculated by fitting a circle to each axial slice. The centers of these circles were then averaged to determine the location of the channel axis. By comparing the center point of each slice to the location of the channel axis, the maximum concentricity can be determined. Similar to cylindrical runout, a perfect cylinder has a concentricity of zero. Concentricity and circular runout were not calculated for the diamond-shaped channel because the final channel cross section was not circular.

Table 2 Geometric data obtained for all coupons

Channel shape	Build orientation	Design D_h (μm)	Measured D_h (μm)	Max cylindrical runout (μm)	Max concentricity (μm)	R_a (μm)	R_a/D_h
Cylindrical	Vertical	508	554	113	34.7	8.40	0.015
Cylindrical	Horizontal	508	569	221	133	16.1	0.028
Cylindrical	Diagonal	508	528	225	133	16.3	0.031
Diamond	Horizontal	508	623	N/A	N/A	13.4	0.022
Teardrop	Horizontal	508	599	214	121	13.5	0.022

Geometric results from these CT scan analyses are provided in Table 2. All of the coupons built had channel hydraulic diameters larger than the design intent. The diagonally built coupon was closest to the design intent, while the diamond-shaped channels had the largest deviation from the design. These deviations are likely due to a combination of nonoptimized process parameters, such as beam offset, hatching density, laser power, and material scaling.

As expected, the vertically built channels had the best concentricity and circularity tolerance, as given by Table 2. This acceptable tolerance is because an exact circle is defined with each layer. In the horizontal and diagonal build directions, however, a noncircular shape is defined with each layer. Moreover, considerably larger geometric errors have been seen in the thickness direction of DMLS parts [6]. This lack of control over the thickness direction, coupled with the layer shapes of the horizontal and diagonal build directions, resulted in higher cylindrical and concentricity errors. The effect of these errors in the horizontally built channels can be seen in Fig. 5. Five axial slices taken through one channel show the variation in channel cross section. Overall, the geometric analysis shows that it is difficult to hold a tight tolerance with DMLS at a small scale. Moreover, the orientation of these small features to the build direction has a significant impact on the attainable tolerance.

The CT scan data were also used to calculate the internal surface roughness of the channels. Roughness measurements required a mean surface to calculate the roughness height. This mean surface was defined by fitting a polynomial surface to the data using the least squares method with bisquare weighting. Figure 6 shows a slice taken through the surface of the vertically built coupon surface, illustrating the roughness features and mean surface. It is noted that the mean surface is not flat, but that the

deviation is quite small at $\pm 20 \mu\text{m}$. After the mean surface was computed, Eq. (2) was used to calculate the surface roughness R_a

$$R_a = \frac{1}{n} \sum_{i=1}^n |z_{\text{surf}} - z_{\text{meas}}| \quad (2)$$

A summary of the roughness values for each coupon is given in Table 2. Roughness levels found for all coupons were similar to those reported for the DMLS process with various materials [1–7]. These roughness levels are more than double than those typically seen with a machining, EDM, or investment casting process [17]. Therefore, using DMLS to create small channels resulted in internal surfaces rougher than traditional methods, regardless of the DMLS build direction or channel shape. At the small scale of these channels, these high roughness levels are significant.

Comparing roughness values for the coupons shows that vertical build direction results in the smoothest interior surface. This is because the interior of the vertical coupon has no downward facing surfaces, which are rougher than upward facing surfaces [18,19]. Downward facing surfaces are surfaces built upon unsintered powder. As the material is sintered, some of the surrounding powder fuses to the downward facing surface, resulting in large ball-like roughness features. This phenomenon can be confirmed qualitatively in the 3D surfaces shown in Fig. 7. The downward facing surface (upper channel wall) is rougher than the upward facing surface (bottom channel wall).

Each having a downward facing surface, one would expect the horizontally built coupons to have the same roughness. However, as shown in Table 2, the teardrop- and diamond-shaped channels are smoother than the cylindrical-shaped channels. This can be explained by the fusing of the small upper surfaces in the diamond and teardrop coupons. This fusing effectively smoothed some of the roughness features that would have been present on the each individual surface.

Experimental Rig

A test rig was built to collect pressure drop and heat transfer measurements of each of the test coupons using a similar design to that already presented by Weaver et al. [20] and Stimpson et al.

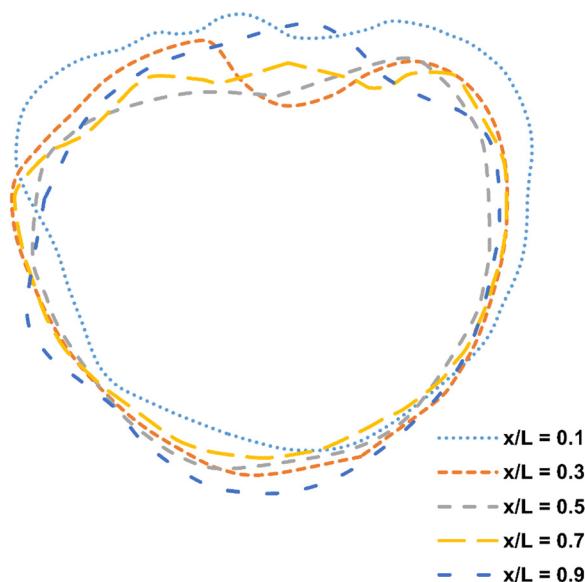


Fig. 5 Axial slices of horizontally built cylindrical channel showing nonuniform channel cross section

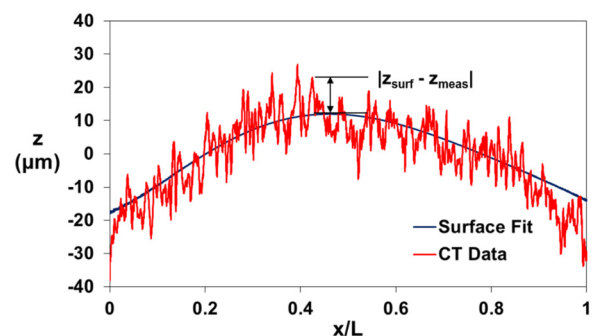


Fig. 6 A streamwise slice through the cylindrical-vertical point cloud showing the fitted surface and roughness elements

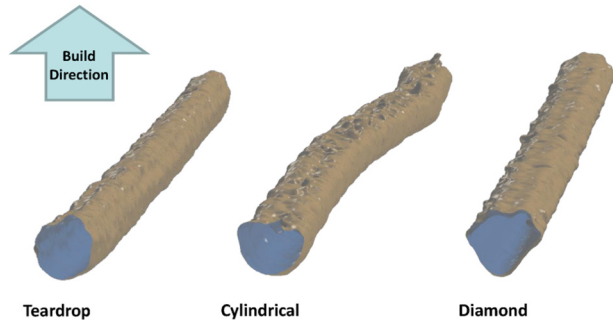


Fig. 7 Three-dimensional tessellated surfaces comparing final channel quality for horizontally built channels

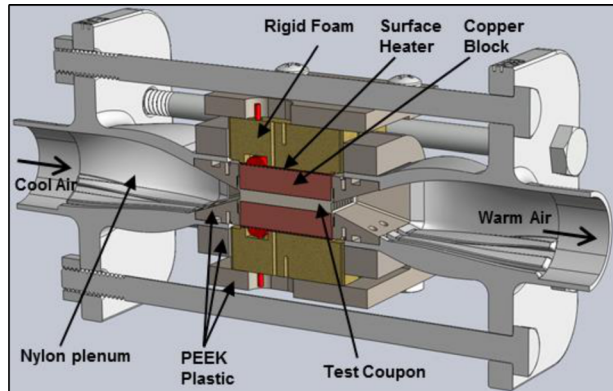


Fig. 8 CAD image of rig used to measure pressure drop and heat transfer in additively manufactured test coupons

[18]. The rig, shown in Fig. 8, was built with a smooth contraction chamber that supplied uniform velocity air to the coupon inlet. The exit expansion chamber was made identical to the inlet for simplicity of fabrication. The coupon was mounted between the inlet and exit pieces and sealed with rubber gaskets between the mating surfaces.

Pressure taps were installed upstream of the inlet contraction and downstream of the exit expansion to measure the pressure drop across the coupon. This pressure drop was modified to account for expansion losses associated with a sharp exit into a large reservoir. Mass flow rate through the rig was measured using a laminar flow element with accompanying pressure and temperature measurements. The pressure drop across the coupon, inlet temperature, static pressure, and mass flow rate were used to calculate the friction factor with the below equation

$$f = \frac{D_h 2 \Delta P}{L \rho v^2} \quad (3)$$

During the heat transfer experiments, heat was added to the upper and lower surfaces of the test coupon using electrical resistance surface heaters. The heaters were adhered to copper blocks with a thin layer of thermally conductive paste. The thickness of the copper blocks was selected to ensure that the temperature along the length of the coupon was uniform to create a constant temperature boundary condition in the coupon. Measured by imbedded thermocouples, the temperature of the copper block was used to calculate the channel surface temperature, T_s , using a 1D conduction analysis through the copper, thermal paste, and coupon wall. Conduction losses through the polyether ether ketone plastic components were accounted for using imbedded thermocouples. Conduction losses were a maximum of 11% for the teardrop coupon at the lowest Reynolds number. However, conduction losses for most cases were 3–6%.

The convective heat transfer coefficient, h , was calculated using Eq. (4), where A_s is the wetted surface area of the channels determined using the CT scan data. Q_{air} represents the heat transferred to the fluid determined by Eq. (5), while ΔT_{LM} is the log mean temperature difference as defined in Eq. (6)

$$h = \frac{Q_{\text{air}}}{A_s \Delta T_{\text{LM}}} \quad (4)$$

$$Q_{\text{air}} = \dot{m} c_p (T_2 - T_1) \quad (5)$$

$$\Delta T_{\text{LM}} = \frac{T_{\text{air},2} - T_{\text{air},1}}{\ln \left(\frac{T_s - T_{\text{air},1}}{T_s - T_{\text{air},2}} \right)} \quad (6)$$

The heat transferred to the air, Q_{air} , should be the same as the heat added to the system from the electric heaters, less the conduction losses. By measuring the power dissipated by the heaters using a precision resistor, there were redundant measurements of heat input to the fluid. For all of the experiments reported in this paper, the difference between these two heat input measurements was less than 10%.

Measurement Uncertainty. Uncertainty of the measurement method described above was determined using the analysis method described by Figliola and Beasley [21]. Multiple pressure transducers were used to optimize measurement resolution throughout the range of Reynolds numbers. Therefore, the percent uncertainty was not constant across each dataset. Nonetheless, the uncertainty for all friction factor measurements was less than 7%. Uncertainty in Nusselt number measurements was also within 7% for all data. Uncertainty in the Reynolds number ranged from 4.5% to 7%.

For friction factor, Nusselt number, and Reynolds number, the largest contributor to uncertainty was the hydraulic diameter. This is displayed graphically for friction factor and Nusselt number in Fig. 9. This uncertainty in hydraulic diameter was 2.5% and was driven by the uncertainty in the geometric data.

As stated in the coupon description section above, multiple coupons of each design were constructed. Upon testing the duplicate coupons, the variations in friction factor and heat transfer were found to be within the measurement uncertainty. No difference was seen between coupons of the same design. Therefore, the data shown in Figs. 10–14 are only from one coupon, but is representative of all three coupons of a given design.

Build Effects on Coupon Performance

Friction factor performance of the cylindrical channels is compared between the different build directions in Fig. 10. This friction factor data were calculated using the measured hydraulic diameters and cross-sectional areas determined with the CT scan. Also shown is rig benchmarking data from Stimpson et al. [18], which shows good agreement with correlations for a smooth channel of similar dimensions.

As was expected based on previously reported data in the literature for rough channels, the laminar data do not fall on the 64/Re

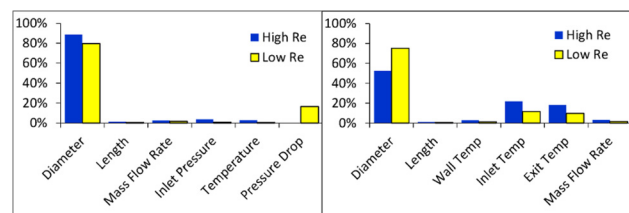


Fig. 9 Relative contributions to uncertainty of friction factor (L) and Nusselt number (R) from dominant measurements

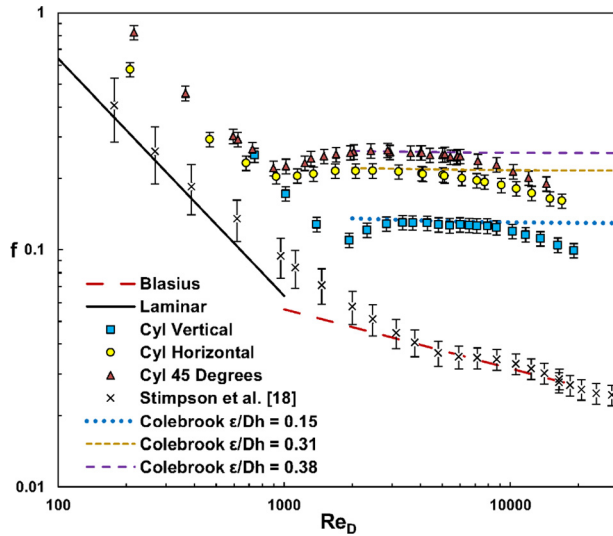


Fig. 10 Cylindrical channel friction factor performance for three different build directions and benchmarking data from Stimpson et al. [18]. Benchmarking error bars range from 10% to 30%, while current study error bars are 7%.

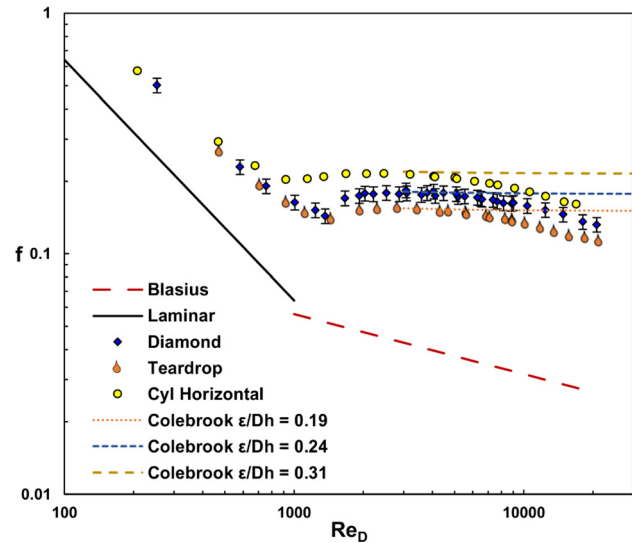


Fig. 12 Friction factor data for horizontally built coupons with Colebrook curves matched to the peak turbulent friction factor

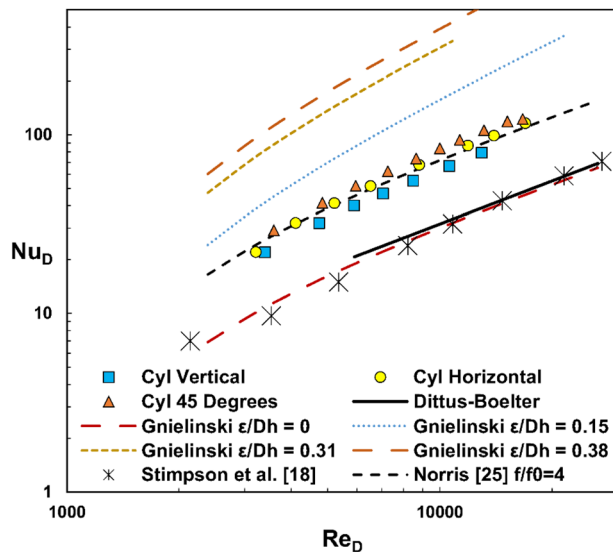


Fig. 11 Heat transfer performance for cylindrical coupons comparing different build directions and benchmarking data from Stimpson et al. [18]. Marker size represents the uncertainty: current study error is 7% and benchmarking data error is 12%.

curve [22,23]. Earlier transition between flow regimes shown in Refs. [22] and [23] was also confirmed with this data. Transition to turbulent flow occurred at $Re \sim 900$ for the horizontal and diagonal channels; the roughest channels were tested. The vertically built channels transitioned at $Re \sim 2000$ and showed the lowest friction factor.

The variation between the friction factors can be explained by examining the relative roughness of the channels in each build direction. Vertically built channels have the smallest relative roughness as shown in Table 2. Following the vertical coupon in relative roughness is the horizontally and diagonally built coupons, which have the same relative roughness and have similar friction factors.

Also plotted in Fig. 10 are fully turbulent curves of the Colebrook equation, as shown in Eq. (7). These curves were generated

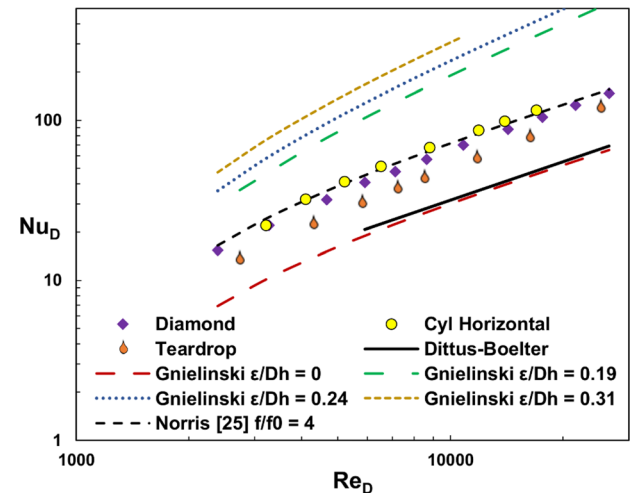


Fig. 13 Heat transfer performance of horizontally built coupons. Curves show Gnielinski correlation using friction factor taken from curves in Fig. 12. The size of the marker represents the uncertainty of 7%.

by manually adjusting the relative roughness value (ϵ/D_h) so that each curve matched one dataset in Fig. 10. The relative roughness values determined from this analysis are much larger than R_a/D_h . However, this is not surprising since the Colebrook equation is based upon a sand grain roughness. Nonetheless, the Colebrook ϵ/D_h is much larger than the typical range plotted on the Moody diagram. This suggests that these channel flows are not behaving as a typical rough channel with moderate roughness. Instead, large roughness features protruded into the flow, acting like small obstructions and driving the pressure loss. These roughness features can be seen in Fig. 7

$$\frac{1}{\sqrt{f}} = -2 \log_{10} \left(\frac{\epsilon}{3.7D_h} + \frac{2.51}{Re\sqrt{f}} \right) \quad (7)$$

Despite the differences in friction factor, the heat transfer performance was similar between the three different build directions. This performance is presented as a Nusselt number, as defined in the “Nomenclature” section. Data for each build direction, as well

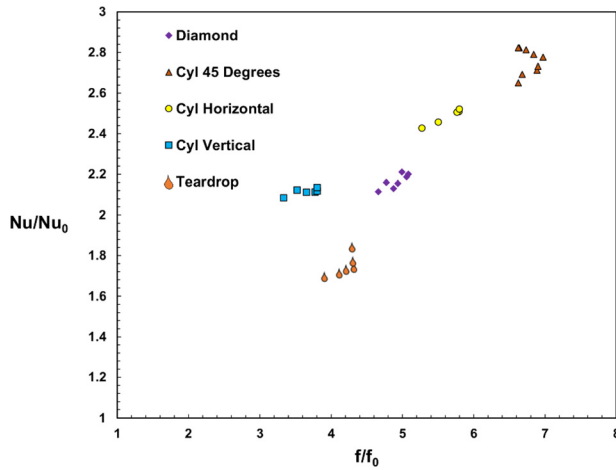


Fig. 14 Friction factor and heat transfer augmentation for all tested channels

as rig benchmarking data from Stimpson et al. [18], are shown in Fig. 11. Uncertainty in Nusselt number data is represented by the size of the data marker. Also given in Fig. 11 are curves representing the Gnielinski [24] correlation, Eq. (8), at different values of relative roughness. These values match those previously deduced from the Colebrook equation

$$Nu_D = \frac{\left(\frac{f}{8}\right) (Re_D - 1000) Pr}{1 + 12.7 \left(\frac{f}{8}\right)^{0.5} (Pr^{2/3} - 1)} \quad (8)$$

The heat transfer data for each coupon clustered well below the predicted Nusselt number using the calculated ε/D_h from the Colebrook equation. Also, even though the horizontal and diagonal coupons had a much higher relative roughness than the vertical coupon, this larger roughness did not cause an appreciable increase in heat transfer.

The effect of increased pressure loss with little increased heat transfer was shown by Norris [25], where heat transfer ceases to increase with roughness when the friction factor augmentation (f/f_0) is larger than 4. Equation (9) was taken from Ref. [25] to calculate a heat transfer curve for $f/f_0 = 4$. Nu_0 was determined using the Gnielinski correlation for a smooth pipe ($\varepsilon/D_h = 0$). The curve generated from the Norris correlation is plotted in Fig. 11, confirming that no data extend beyond the Nusselt number for $f/f_0 = 4$. By referring to augmentation data shown in Fig. 14, this result is corroborated since the friction factor augmentation of most data for this study was larger than 4

$$Nu = Nu_0 \left(\frac{f}{f_0}\right)^n \quad n = 0.068 \quad Pr^{0.215} \quad (9)$$

Alternate Channel Geometry Performance

Comparing friction factor of all the horizontally built coupon shapes in Fig. 12, the cylindrical channel shape transitioned to turbulent flow at $Re \sim 900$, while the teardrop and diamond channels transitioned at $Re \sim 1500$. This decrease in transitional Reynolds number with increasing roughness is consistent with the literature [22,23].

As with the coupons built in different directions, the differences in friction factor between channel shapes can be explained by their relative roughness. The cylindrical-shaped channels had the highest relative roughness and highest friction factor of the horizontal coupons. Both the diamond and teardrop shapes have lower friction factor and relative roughness than the cylindrical shape.

Even though the diamond shape appears to have a slightly higher friction factor than the teardrop shape, this difference is within the measurement uncertainty.

Figure 7 presented 3D reconstructions of the horizontal coupons, which can be used to qualitatively examine the roughness features affecting the flow. The top of the cylindrical channel was unsupported during the build, resulting in large peaks and valleys. These large roughness features can be seen in Fig. 7 protruding into the channel causing the high pressure loss.

Similar to the coupons built in different directions, the heat transfer performance of the horizontally built coupons varied little from coupon to coupon. As shown in Fig. 13, Nusselt numbers for the three coupon shapes were similar; however, there are slight differences between the data. These differences match the friction factor trends shown in Fig. 12. The smoothest coupon, which was the teardrop shape, has the lowest heat transfer. The cylindrical-shaped channel, on the other hand, was the roughest and had the highest heat transfer. Comparing these surfaces in Fig. 7, the upper sides of the teardrop shape have fused together to form a smoother surface. Without the large roughness peaks, there are less features to promote turbulence and increase heat transfer.

Augmentation Results

To evaluate the relative increases in friction factor and Nusselt number for the different tests, augmentation values were calculated. For friction factor, f_0 was calculated using Eq. (7), the Colebrook equation for flow through a smooth pipe. For Nusselt number, the baseline Nu_0 was determined using Eq. (8), the Gnielinski correlation for a smooth pipe. The augmentation values for all of the coupons are shown in Fig. 14. Showing the lowest friction factor augmentation were the vertically built cylindrical channels, while the cylindrical channels built diagonally showed the highest heat transfer augmentation.

Looking at the grouping of points in Fig. 14, the vertically built channels lie separate from the trend of the other coupons. This is likely due to the difference in type of roughness features present. Since the vertical coupon had no downward facing surfaces, the roughness features are of different form. Adding circular layers to build the vertical coupon resulted in riblike roughness features. These roughness features are more effective at augmenting heat transfer without penalizing friction factor.

Since channel flow in gas turbines is typically driven by a fixed pressure ratio, it is useful to evaluate the reduction in mass flow from a smooth channel to a rough DMLS channel. The fixed pressure ratio for the analysis was taken from the cylindrical diagonal dataset at $Re = 10,000$. This coupon was chosen for this analysis because it had the highest friction factor, and therefore, the largest mass flow reduction. Comparing this DMLS channel to a theoretical smooth channel showed a 67% reduction in mass flow rate at the given pressure ratio. The resulting heat transfer from this reduction in mass flow was also calculated. The result showed that the heat transfer for the DMLS coupon was 40% lower than the smooth coupon, despite the mass flow decreasing by 67%. Therefore, the augmentation of the heat transfer due to the roughness partially compensates for the decrease in the heat transfer from the reduced mass flow.

Conclusions

Coupons for heat transfer and flow testing were manufactured using DMLS. Cylindrical-shaped channels were built in three different orientations, while teardrop and diamond channels were built horizontally. A CT scanner was used to characterize the dimensions of the as-built coupons. Significant deviations from the design intent were found in all but one coupon. Additionally, the CT scan was used to determine internal surface roughness. Through this analysis, the vertically built coupons were determined to have the lowest surface roughness. However, all coupons exhibited an internal surface roughness much greater than traditional manufacturing techniques.

Flow testing the coupons showed that these vertically built coupons had the lowest friction factor, while the diagonally built coupons had the highest friction factor. These differences were attributed to geometric errors in the layers. Of the alternate channel shapes, the teardrop had the lowest friction factor while allowing the channel to retain a cylindrical form in the horizontal build direction. This was accomplished when the two small upper surfaces of the teardrop shape fused together during the build process.

These large differences in friction factor between build orientation and channel shape did not result in similar differences in Nusselt number. Heat transfer performance was comparable across all coupons. From a gas turbine designer's perspective, at a fixed pressure ratio the rough DMLS surface helps to recover heat transfer lost due to the drop in mass flow from a theoretically smooth channel. If better DMLS tolerances can be held for channels at engine scale, then the inherently rough surface of DMLS could be used to create highly efficient cooling geometries.

Ultimately, build direction had an appreciable effect on the internal features of DMLS parts. While AM opens the design space for gas turbine cooling, variables in the DMLS process must be considered in order to produce high-quality parts.

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Nomenclature

A_c = cross-sectional area
 A_s = wetted surface area
 c_p = specific heat capacity of air
 D_h = hydraulic diameter ($4A_c/p$)
 f = Darcy friction factor
 h = heat transfer coefficient
 k = thermal conductivity
 L = channel length
 m = number of channels
 n = number of point cloud points
 Nu_D = Nusselt number (hD_h / k_{air})
 p = perimeter
 P = pressure
 Pr = Prandtl number
 Q_{air} = heat transferred to the fluid
 R_a = mean surface roughness
 T = temperature
 v = fluid velocity
 x = streamwise position
 z = roughness height
 ΔP = pressure differential

Greek Symbols

ε = sand grain roughness
 ρ = air density

Subscripts

meas = CT scan measured part surface
 s = surface at interface of fluid and solid

0 = smooth channel baseline
 1 = inlet
 2 = exit

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